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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2024
END SEMESTER EXAMINATION - ODD 2024
CS101 - PROGRAMMING FOR PROBLEM SOLVING

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) If $x = 20$, $y = 4$, then what is the value of z where $z = x++ + y$. (1) CO1 BL2

- a) 20
- b) 25
- c) 24
- d) 30

1.ii) Which pair of functions below are used for single character I/O. (1) CO3 BL3

- a) getchar() and putchar()
- b) scanf() and printf()
- c) Input() and output()
- d) None of these

1.iii) What is the output of the following code – (1) CO2 BL2

```
#include <stdio.h>
int main() {
    int i = 2;
    switch (i) {
        case 0:

            printf("Hello");
            break;
        case 1:
            printf("World");
            break;
        default:
            printf("HelloWorld");
    }
    return 0;
```

- a) Hello
- b) World

- c) HelloWorld
- d) Error

1.iv) Maximum number of elements in the array declaration `int a [5][8]` is (1) CO3 BL4

- a) 28
- b) 32
- c) 35
- d) 40

1.v) Obtain the 2's complement for 11011. (1) CO1 BL3

- a) 00110
- b) 10110
- c) 11011
- d) 00101

1.vi) What will be the output of the following C code? (1) CO3 BL2

```
#include <stdio.h>
int main()
{
    int i = 0;
    do {
        i++;
        printf("In while loop\n");
    } while (i < 3);
}
```

- In while loop
- In while loop
- a) In while loop
- In while loop
- b) In while loop
- c) Depends on the compiler
- d) Compile time error

1.vii) The Hexadecimal equivalent of the number (101101010010) 2 is (1) CO5 BL4

- a) A52
- b) B52
- c) A54
- d) C52

1.viii) Which of the options is correct, if we run the following program: (1) CO5 BL4

```
#include<stdio.h>
```

```
void main()
```

```
{
```

```
    int num[3]={10,20,30}, *ptr;
```

```
    ptr = &num[0];
```

```
printf("%d",*(ptr+3));

}
```

- a) The output is : 30
- b) The output is: 20
- c) Compiler time error is generated
- d) None of these

1.ix) What is a NULL pointer in C?

(1) CO4 BL1

- a) A pointer that points to a random memory address
- b) A pointer that points to the address of a variable
- c) A pointer that does not have a valid memory address assigned
- d) A pointer that points to the address of another pointer

1.x) What is default storage class of variables in C language?

(1) CO4 BL3

- a) extern
- b) local
- c) auto
- d) global

1.xi) What is the output of this C code?

(1) CO5 BL4

```
# include < stdio.h >

void main()

{

    float k=0;

    for(k=0.0;k<3.0;k++);

    printf("%f",k);

}
```

- a) 2.00
- b) 3.00
- c) 4.00
- d) None of these

1.xii) What is the output of the following code? for(int i = 0; i < 5; i++) { if(i == 3) continue; printf("%d ", i); }?

(1) CO3 BL3

- a) 0 1 2 3 4
- b) 0 1 2 4
- c) 1 2 3 4
- d) 0 1 2

Group B

(Short Answer Type Questions)

(Answer any three of the following) 3x5=15

2. What is call by value and call by reference? Explain with example.

(5) CO4 BL3

- | | | | |
|--|-----|-----|-----|
| 3. Explain the difference between an 'entry control loop' and an 'exit control loop' with example | (5) | CO5 | BL4 |
| 4. Write a parameterized macro to calculate the square of an integer. Now use this macro in main() function to calculate square of an integer taken from the user. | (5) | CO3 | BL3 |
| 5. Write a c program to add the diagonal elements of a 2D square array. | (5) | CO5 | BL4 |
| 6. Write a C program to find the maximum difference between two elements in an array. | (5) | CO2 | BL3 |

Group C

(Long Answer Type Questions)

(Answer any three of the following) 3x15=45

- | | | | |
|--|------|-----|-----|
| 7. Answer the following questions. | (15) | | |
| a) Convert the following:
(752.37) ₈ = (?) ₁₆

(37.25) ₁₀ = (?) ₂

(1101110101) ₂ = (?) ₈

Find 2's Complement of (11000.11) | (8) | CO1 | BL4 |
| b) 10110 - 11010 using 2'S Complement | (4) | CO1 | BL4 |
| c) Write a C program to find the maximum number among three numbers using ternary operator. | (3) | CO5 | BL4 |
| 8. Answer the all questions. | (15) | | |
| a) State the difference between implicit and explicit type casting? Explain each with example. | (3) | CO2 | BL2 |
| b) Differentiate between structure and union. | (4) | CO2 | BL3 |
| c) Create a structure to specify data on students given below:
Roll Number, Name, Department, Course, and Year of admission. Assume that there are not more than 450 students in collage. Write a program to print names of all students who joined in a particular year. | (8) | CO5 | BL4 |
| 9. Answer the all questions. | (15) | | |
| a) What is the difference between auto and static variables? Explain with example | (5) | CO3 | BL3 |
| b) Write a C program to calculate the factorial value a number n using recursion. | (5) | CO4 | BL4 |
| c) Write a C program to find the Transpose a matrix. | (5) | CO5 | BL4 |
| 10. Answer the all questions. | (15) | | |

- a) Write a C program to generate following pattern . (6) CO5 BL4
ABCDEFGH
ABCDEFG
ABCDE
ABCD
ABC
AB
A
- b) Write a C program to check whether a year input from keyboard is leap year or not. (5) CO3 BL4
- c) Explain the working principal of strlen(),strcat(),strcpy() and strcmp(). (4) CO3 BL3
11. Write short notes on any three of the following: 3 x 5 (15) CO5 BL4
- a) Dynamic allocation of the memory
 - b) Bitwise Operator
 - c) Pointer Arithmetic
 - d) Functions of Memory unit of a Digital Computer
 - e) Array of Structure
 - f) Pointer to Pointer.

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